

Homework (Jalapeno)

- Q1.** The quadratic function $f(x) = a(x - h)^2 + k$ is in vertex form. What does (h, k) represent?
- (A) The x- and y-intercepts
(B) The vertex of the parabola
(C) The roots of the equation
(D) The maximum and minimum values
(E) The axis of symmetry
- Q2.** The ancient Egyptians used quadratic equations to build the Great Pyramid. The equation $f(x) = x^2 + 5x + 6$ can be written in intercept form as:
- (A) $f(x) = (x + 2)(x + 3)$
(B) $f(x) = (x - 2)(x - 3)$
(C) $f(x) = (x + 1)(x + 6)$
(D) $f(x) = (x - 1)(x - 6)$
(E) $f(x) = (x + 3)(x + 2)$
- Q3.** The quadratic function $f(x) = (x - 2)(x + 4)$ is in intercept form. What is the vertex of the parabola?
- (A) $(-1, -3)$
(B) $(1, -3)$
(C) $(-2, 0)$
(D) $(2, 0)$
(E) $(1, 3)$
- Q4.** In ancient Greece, mathematicians used quadratic equations to model the trajectory of projectiles. The equation $f(x) = x^2 - 4x - 5$ can be written in vertex form as:
- (A) $f(x) = (x - 2)^2 - 9$
(B) $f(x) = (x + 2)^2 - 9$
(C) $f(x) = (x - 1)^2 - 6$
(D) $f(x) = (x + 1)^2 - 6$
(E) $f(x) = (x - 3)^2 - 14$
- Q5.** The Mayans used quadratic equations to model population growth. The equation $f(x) = (x + 1)(x - 3)$ can be written in vertex form as:
- (A) $f(x) = (x - 1)^2 - 4$
(B) $f(x) = (x + 1)^2 - 4$
(C) $f(x) = (x - 2)^2 - 2$
(D) $f(x) = (x + 2)^2 - 2$
(E) $f(x) = (x - 1)^2 - 2$
- Q6.** The quadratic function $f(x) = x^2 + 2x + 1$ is in the form $f(x) = a(x - h)^2 + k$. What is the value of (h, k) ?
- (A) $(0, 0)$
(B) $(0, 1)$
(C) $(1, 0)$
(D) $(-1, 0)$
(E) $(-1, 1)$
- Q7.** The Rosetta Stone, discovered in ancient Egypt, contains quadratic equations. The equation $f(x) = (x + 2)(x - 2)$ can be written in vertex form as:
- (A) $f(x) = (x - 2)^2 - 4$
(B) $f(x) = (x + 2)^2 - 4$
(C) $f(x) = x^2 - 4$
(D) $f(x) = x^2 + 4$
(E) $f(x) = x^2 - 2$
- Q8.** In ancient Rome, engineers used quadratic equations to design aqueducts. The equation $f(x) = x^2 - 3x - 4$ can be written in intercept form as:
- (A) $f(x) = (x - 4)(x + 1)$
(B) $f(x) = (x + 4)(x - 1)$
(C) $f(x) = (x - 1)(x + 4)$
(D) $f(x) = (x + 1)(x - 4)$
(E) $f(x) = (x + 2)(x - 2)$

Open Ended Questions (FRQ)

- Q9.** Write the quadratic function $f(x) = x^2 + 5x + 6$ in vertex form and find its vertex. Explain your reasoning in 5 lines or less.
- Q10.** Write the quadratic function $f(x) = (x - 2)(x + 4)$ in vertex form. Find its axis of symmetry and explain why it is an important feature of the parabola. Provide your answer in 5 lines or less.

Chef's Tips

- Tip 1: Make sure students show work for FRQs
- Tip 2: Review historical contexts to ensure accurate and unique scenarios
- Tip 3: Emphasize the importance of understanding vertex and intercept forms